Absorption and storage

Vitamin A and carotene are absorbed from the small intestine into the lymphatic system. The maximum absorption is reached after 3 to 5 hours of food consumption. The rate of absorption of vitamin A is more rapid than that of carotene. In the adult healthy human only 3% of β-carotene is converted into vitamin A. About 95% of the vitamin A stored in our body is found in the liver and small amount is present in the lungs, adipose tissue and kidneys.

DO YOU KNOW?

Cooking and mashing the vegetables, is it good or bad for β-carotene absorption?

We increase the absorption of β-carotene by steaming, juicing or mashing vegetables. We have to chew well the raw vegetables and fruits in order to maximise the absorption. This causes the cell membranes to rupture and releases more carotene for absorption.

Also the raw vegetables and fruits contain β-carotene in all-trans configuration and cooked vegetables and fruits contain cis-configuration of β-carotene which are more stable and more bioavailable. Therefore, it is best to cook vegetables and fruits in order to maixmise the β-carotene absorption.

Deficiency:

The earliest sign of vitamin A deficiency is concerned with vision. Initially, there is a loss of sensitivity to green light, followed by impairment to adapt to dim light. This condition leads to night blindness. On prolonged or severe deficiency, ulceration of cornea occurs and this condition is known as xerophthalmia or keratomalacia (Figure 8.3).

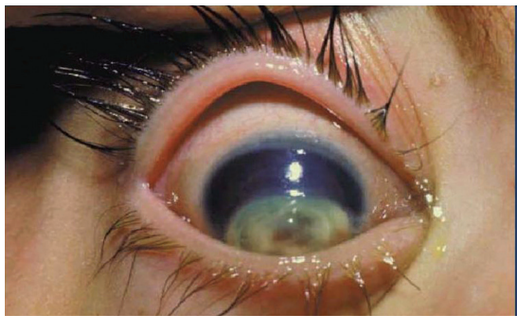


Figure 8.3 keratomalacia.

8.1.2 Vitamin D:

Vitamin D is a sterol compound. It is represented by a group of steroids occuring chiefly in animals, plants and yeast. There are two distinct forms of Vitamin D, namely Vitamin D_2 and Vitamin D_3. Vitamin D_3 is more active than D_2.

Vitamin D_2:

Vitamin D_2 is also known as ergocalciferol. It is produced by the exposure of ergosterol to UV radiation. Ergosterol occurs in plants, milk and yeast.

Vitamin D_3:

Vitamin D_3 is also known as cholecalciferol. It is the natural form of the vitamin ingested from food. It can also be formed by the exposure of 7-dehydrocholesterol present in the skin to sunlight (UV radiation).

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